higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

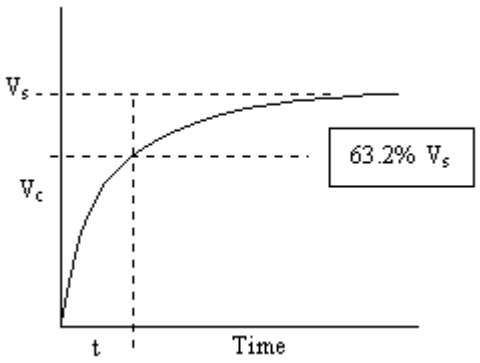
04 April 2023

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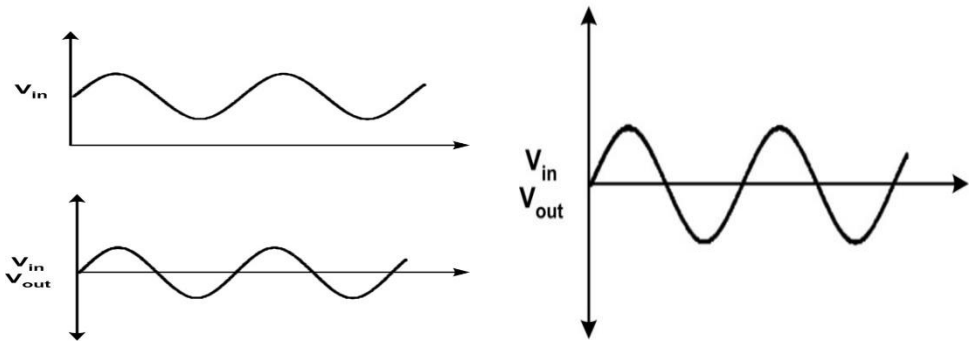
QUESTION 1: ALTERNATING CURRENT THEORY

1.1.1



Graph: **1 mark** for curve; **½ mark** for the V_{in} and time (t) labelling respectively

1.1.2



Graph: **1 mark** for input waveform; **1 mark** for output waveform

(2 × 2) (4)

1.2 Blocks DC signals✓
Allows AC signal to pass✓ (2)

1.3 Square waveforms are easily disturbed by unwanted circuit conditions, hence are used for testing circuits. (2)

1.4 $f_r = \frac{1}{2\pi\sqrt{LC}}$

$C = \frac{1}{(2\pi f)^2 L}$✓

$C = \frac{1}{(2\pi\times1000)^2\times40\times10^{-3}}$✓

$= 0.633\ \mu F$ ✓ (3)

[11]

QUESTION 2: POWER SUPPLIES

2.1 2.1.1

$$I_Z = \frac{P_Z}{V_Z}$$

$$I_Z = \frac{600 \times 10^{-3}}{10}$$

$$= 60\ m\ A\checkmark$$

$$= \frac{V_{in(\min\ )} - V_Z}{I_Z}$$

$$= \frac{16-10}{0.06}$$

$$= 100\ \Omega\checkmark \quad (4)$$

2.1.2

$$R_{L(\min\ )} = \frac{V_z}{V_{in(\min\ )} - V_z} \times R_s$$

$$= \frac{10}{16-10} \times 100\checkmark$$
$$= 166.67\Omega\checkmark \quad (2)$$

2.2 2.2.1

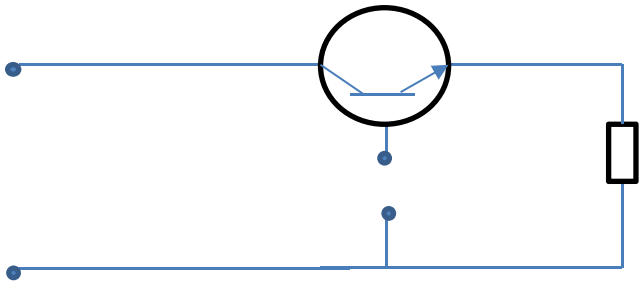
1: INPUT
2: GROUND/GND/COMMON
3: OUTPUT

2.2.2

1: INPUT✓
2: OUTPUT✓
3: GROUND/GND/COMMON ✓

(3 × 2) (6)

2.3



(4)

2.4

- Series voltage regulator with amplifier
- Shunt voltage regulator
- Series voltage regulator with adjustable reference source (Any 2 × 2) (4)

[18]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 3.1.1 $I_e = I_c = 10$ mA

$\therefore V_e = \frac{1}{10} \times V_{CC}$

$= \frac{1}{10} \times 12$ ✓

$= 1,2$ V✓

$R_C = \frac{V_{cc} - V_{ce} - V_e}{I_c}$

$= \frac{12 - 6 - 1,2}{10 \times 10^{-3}}$ ✓

$= 480\ \Omega$✓ (4)

3.1.2 $R_{b2} = \frac{1}{10} \times \beta R_e$

$= \frac{1}{10} \times 250 \times 120$ ✓

$= 3\,000\ \Omega$✓

$V_b = V_e + V_{be}$

$= 1,2 + 0,6$✓

$= 1,8$ V✓

$Rb1 = \frac{R_{b2}(V_{cc} - V_b)}{3\,000(12 - 1,8}{1,8}$ ✓

$= 17\ k\Omega$✓ (6)

3.2 $Zi = Rb \parallel Z1$

$= hie$

$= 1,2\ k\Omega$✓

$Z2 = \frac{1}{h_{oe}}$

$= \frac{1}{20 \times 10^{-6}}$

$= 50\ k\Omega$✓

$\frac{1}{Z_0} = \frac{1}{R_c} + \frac{1}{Z_2}$ OR $Z_0 = \frac{R_c \times Z_2}{R_c + Z_2}$

$= \frac{1}{480} + \frac{1}{50 \times 10^3}$ ✓ $= \frac{480 \times 50\,000}{480 + 50\,000}$ ✓

$\therefore Z0 =$ **475,436 Ω**✓ **= 475,436 Ω**✓

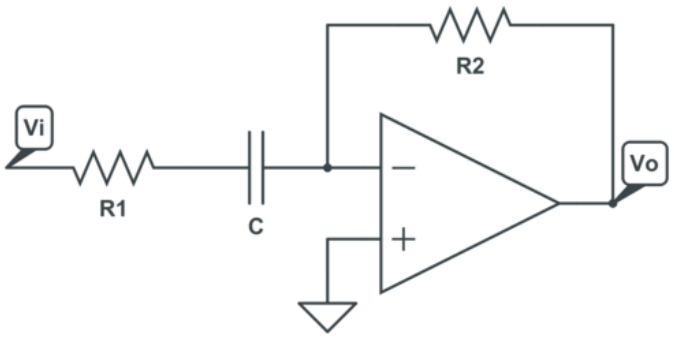
(4)

3.3 3.3.1 Collector current, I_{co}, doubles for each 10^0 in temperature.
3.3.2 Base-emitter voltage decreases with 2,5 mV/0C.

(2 × 1) (2)
[16]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 Practical differentiator



Circuit diagram: **1 full mark** for the op amp; **½ mark** for each resistor, **½ mark each for the input and output.**

4.2 $\tau = \frac{1}{f}$ ✓

$\tau = \frac{1}{500} = 0.002\ s$ ✓

$\tau = RC$ ✓

$C = \frac{\tau}{R} = \frac{0.002}{2000} = 1\mu F$ ✓

(4)
[8]

QUESTION 5: INTEGRATED CIRCUITS

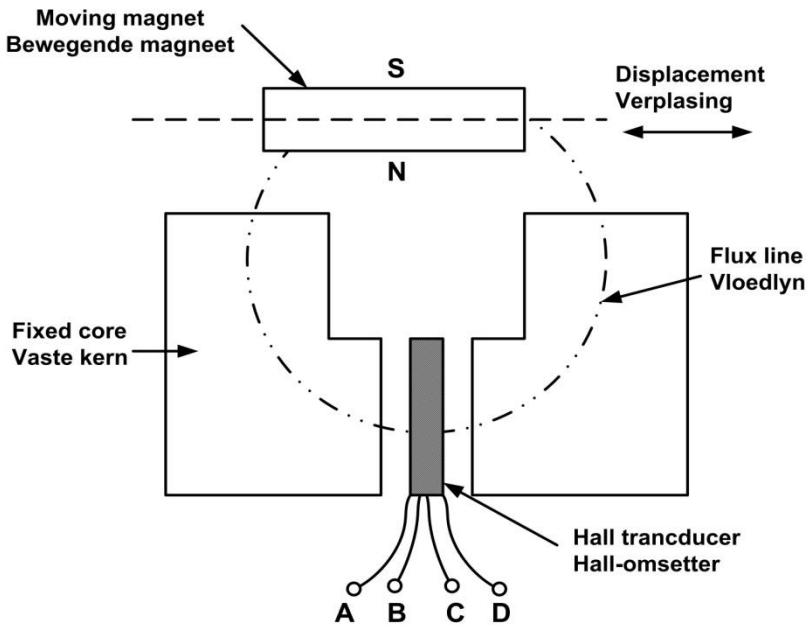
5.1 All unutilised inputs must be connected to V_{DD} **or** V_{DD} (2)

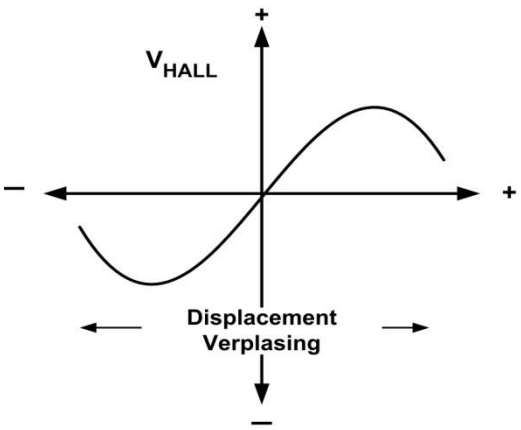
5.2 5.2.1 True
5.2.2 False
5.2.3 True

(3 × 1) (3)
[5]

QUESTION 6: ELECTRONIC PHASE CONTROL

6.1



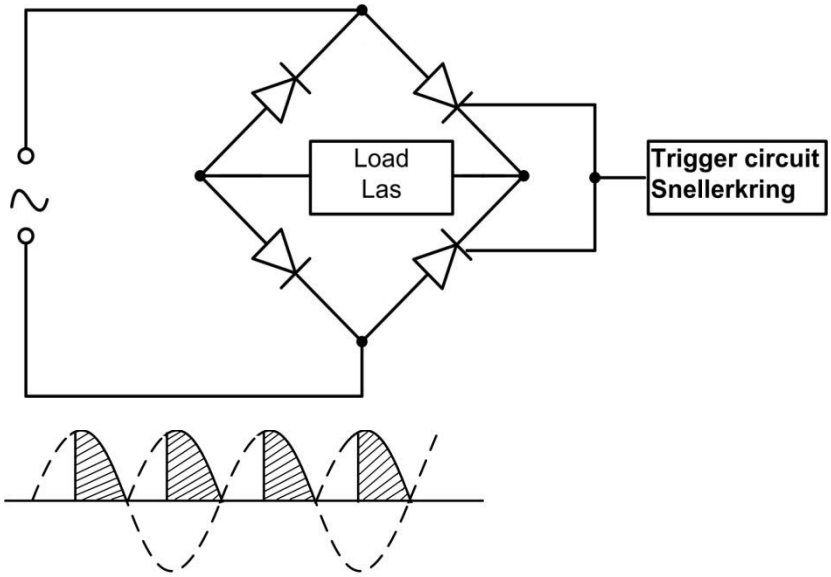


Circuit diagram: ½ **mark** for each correct label on and waveform sketch (6)

6.2
- The capacitor C1 passes the pulse but blocks the slow variation of surrounding light.✓
- The photodiode conveys the pulses to the amplifier. C2 faces the pulses to the smooth direct current which helps T1 on and lets current flow through the relay.✓✓
- As soon as emission of an object or person is interrupted, no signal is conveyed to the transistor and therefore switches it off.✓✓
- The relay switches to its normal condition so that the contacts short circuit, which activates the alarm.✓

(6)
[12]

QUESTION 7: TEST EQUIPMENT



Full-wave DC-control with tow SCRs and two diodes in a bridge

Circuit diagram: **1 full mark** for the correct waveform and trigger circuit label; **½ mark** for the other remaining correct component and labels **[5]**

QUESTION 8: OSCILLATORS

8.1
- The phase shift network of a Hartley oscillator consists of two inductors and a capacitor
- The phase shift network of a Colpitts oscillator consists of two capacitors and an inductor

(4)

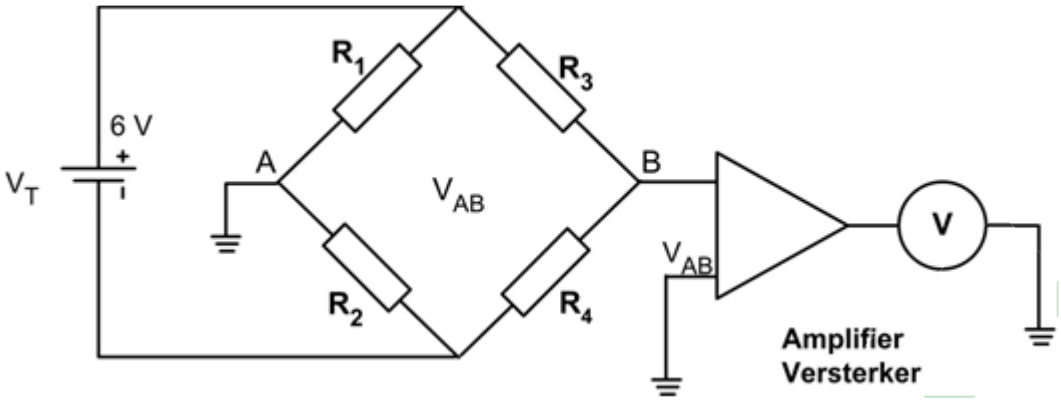
8.2
- The RC phase shift network consists of three identical RC sections and is connected in the feedback line of the capacitor.
- Every RC section causes a phase shift of 600 which means the total phase shift is 1 800.
- Because the transistor supplies a further phase shift of 1800, the total phase shift is 3 600 which means that the input and output waveforms are in phase.

(3 × 2)

(6)
[10]

QUESTION 9: TRANSDUCERS

9.1



THERMISTOR BRIDGE WITH A DC-BIAS (4)

9.2 $T = {}^{0}C + 273$

$= 30 + 273$

$= \underline{\mathbf{303\ K}}$ ✓

9.2.1 $R_t = A \cdot e^{\frac{B}{T}}$

$= 0{,}2169\ e^{\frac{3200}{303}}$

$= \underline{\mathbf{8\ 372{,}747\ \Omega}}$ ✓ (2)

9.2.2 $V_{AB} = V_A - V_B$

$$V_A = \frac{R_2}{R_1 + R_2} \times V_T$$

$$= \frac{10\ k\Omega}{10\ k\Omega + 10\ k\Omega} \times 10$$

$$= \mathbf{5\ V}\checkmark$$

$$V_B = \frac{R_t}{R_t + R_3}$$

$$= \frac{8\,372{,}747}{8\,372{,}747 + 10\,000} \times 10$$

$$= 4{,}56\ V\checkmark\checkmark$$

$$V_{AB} = V_A - V_B$$

$$= 5 - 4{,}56$$

$$= 0{,}44\ V\checkmark\checkmark$$

$$\text{Gain of amplifier (AV)} = \frac{V_O}{V_i}$$

$$= \frac{10}{0{,}44}$$

$$= 22{,}73\checkmark\checkmark \quad (7)$$

[13]

TOTAL: 100

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